Paper Code : A

SUNBEAM INSTITUTE OF INFORMATION TECHNOLOGY
Pune, Karad

Sample Test Paper

Max. Marks: 50

PRN:_____ **Name:** _____ **Batch:** _____

Instructions :

1. Create **new workspace (folder)** by the name **OOP_Batch_12 digit PRN** in your home directory.
2. Create **new Project** with name **OOP_12 digit PRN**.
3. First, compress/zip your folder (named OOP_Batch_12 digit PRN) and upload it on EMIS.

Evaluation of Lab Exam should be based on the following criteria :				
Paper Code – A				
Sr. No.	**Points to cover**	**Tick Mark by student**	**Max Marks**	**Marks by Evaluator**
1.	Classes + Menu Driven		2+2+2+4	
2.	Add Employee		7	
3.	Display All Employees		5	
4.	Search Employee by ID		5	
5.	Display Employees joined in given Year		6	
6.	Find Employee with Maximum Salary		6	
7.	Find Employee with Minimum Salary		6	
8.	Exit the Application		5	
		Total :	**50**	_____

_____ _____

Signature of Student **Signature of Evaluator**

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Question :
Title: ***Employee Management System – Menu Driven Console-Based Java Application using Arrays.***

Scenario:
You are tasked with designing and developing a **menu-driven, console-based Java application** to manage employee records in a company. The application must use **arrays** to store and manipulate employee data. Additionally, the Employee class should have a **date of joining** field and Person class should have a **date of birth** field, implemented using a separate MyDate class (not java.util.Date).
Apply Inheritance wherever necessary in class design.

Class Design Guidelines:
A) Person Class:
I) String **name,** MyDate **dateOfBirth** (association).
II) Constructor, getters, setters, and toString() method.

B) Employee Class:
I) int ***id***, String ***department***, double ***salary***, MyDate ***dateOfJoining*** (association).
II) Constructor, getters, setters, and toString() method.

C) MyDate Class:
I) int ***day***, int ***month***, int ***year.***
II) Constructor, getters, setters, and method toString() to return date in "dd-mm-yyyy" format.

Requirements:
Design and implement a menu program that supports the following operations:
1) Add Employee:
- Allow user to input employee details: ***name*** *(String),* ***dateOfBirth*** *(Date),* ***id*** *(int)* ***department*** *(String),* ***salary*** *(double),* ***dateOfJoining*** *(Date)*
- *day, month, year – to be stored as a* ***MyDate class object****.*
- Use an array of Employee references with a fixed size (e.g., max 10 employees).

2) Display All Employees:
- Show details of all employees which are **currently stored** in the array, including formatted date of joining and date of birth.

3) Search Employee by ID:
- Prompt the user to enter an id and display the matching employee if found.

4) Display Employees joined in given Year:
- Prompt the user to enter a year and display all employees joined in that year.

5) Find Employee with Maximum Salary:
- Traverse the array to find and display the employee with the highest salary.

6) Find Employee with Minimum Salary:
- Traverse the array to find and display the employee with the lowest salary.

7) Exit the Application.

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